

The Master Hardware Map (ESP32-S3 N16R8)

1. The Wake-Up Sensors (Light Sleep Triggers)

- **MQ2 Gas Sensor (D0) → GPIO 1** (*CRITICAL: Requires 2k/3.3k voltage divider to step 5V down to 3.3V*)
- **KY-037 Audio Sensor (D0) → GPIO 2**

2. The I2S Digital Microphone (INMP441)

- **SCK / BCLK → GPIO 4**
- **WS / LRC → GPIO 5**
- **SD / DIN → GPIO 6**
- **L/R → GND** (*Sets microphone to Left Channel*)

3. The I2C Bus (DS3231 Real-Time Clock)

- **SDA → GPIO 8**
- **SCL → GPIO 9**
- **VCC → 3.3V** (*Do not use 5V if a non-rechargeable CR2032 battery is installed*)

4. SPI Bus A: The SD Card Module (FSPI)

- **CS (Chip Select) → GPIO 10**
- **MOSI → GPIO 11**
- **SCK → GPIO 12**
- **MISO → GPIO 13**

5. SPI Bus B: The LoRa Radio (HSPI / SX1278)

- **MOSI → GPIO 14**
- **MISO → GPIO 15**
- **SCK → GPIO 16**
- **CS (NSS) → GPIO 17**
- **RST (Reset) → GPIO 18**
- **DIO0 (Interrupt) → GPIO 7**